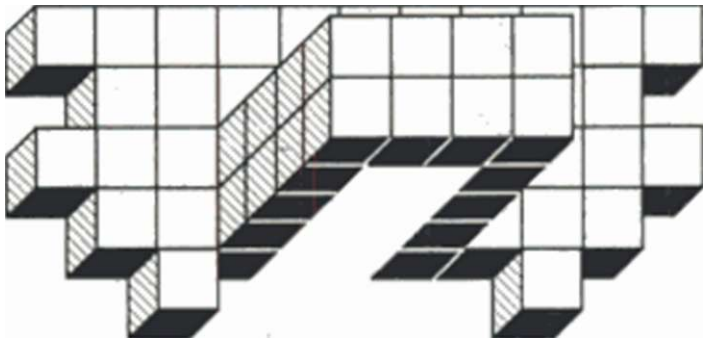


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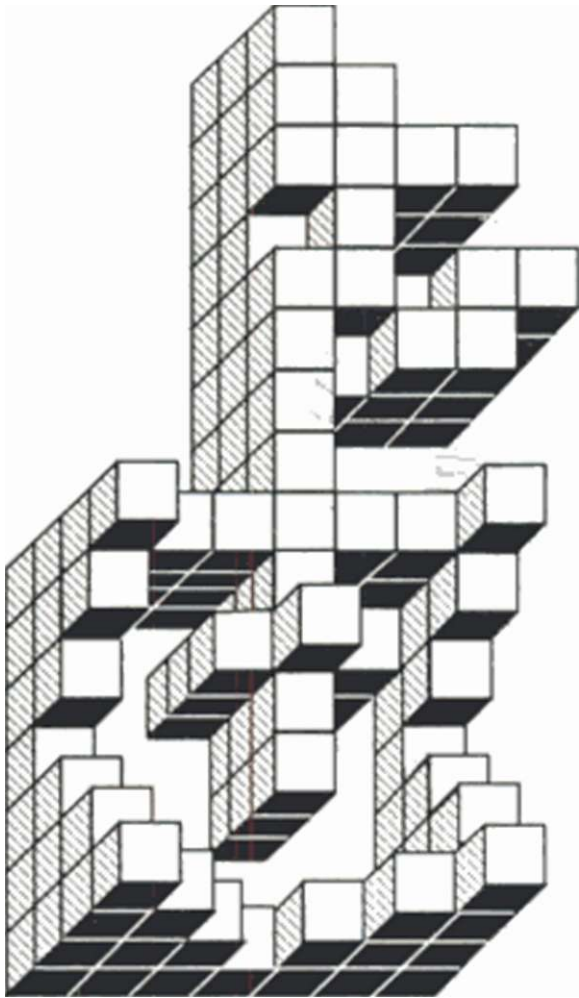


- (a) 57

(c) 60
- (ft) 58

(d) 62

12.



- (a) 130

(c) 138
- (b) 132

(d) 140

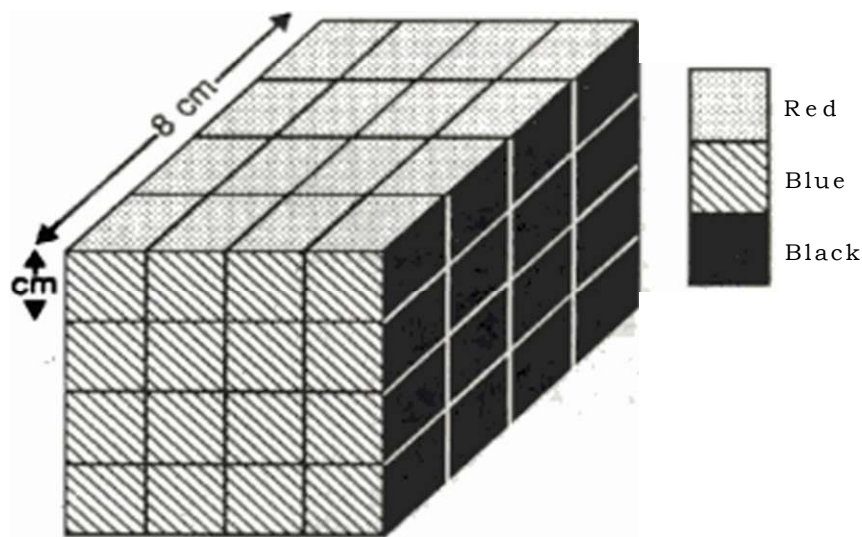
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- 1.(d): There are four cubes in the middle layer which have one face painted only in blue.
- 2.(6): There is one (central) cube in the top layer and one (central) cube in the bottom layer which have one face painted only in Green.
- 3.(c): There are 9 cubes in each of the three layers. Thus there are 27 cubes in all.
- 4.(a): Four (corner) cubes in the top layer and four (corner) cubes in the bottom layer have three sides painted. Hence, there are 8 such cubes.
- 5.(e): Only one central cube in the middle layer has no faces painted at all.

EXERCISE 14B

- Q. 1. Directions : A cube painted yellow on all facet is cut into 27 small cubes of equal sizes. Answer the questions that follow :
- (1) How many cubes are painted on one face only ?
(o) 1 (6) 6 (c) 8 (<f) 12
 - (2) How many cubes are not painted on any face ?
(a) 1 (b) 4 (c) 6 (d) 8
- Q. 2. All surfaces of a cube are coloured. If a number of smaller cubes are taken out from it, each side 1/4th size of the original cube's side, indicate the number of cubes with only one side painted :
- (a) 16 (6) 20 (c) 24 (d) 40
- Q. 3. Direction* : A solid cube of each side 8 cms, has been painted red, blue and black on pairs of opposite faces. It is then cut into cubical blocks of each side 2 cms.



- (1) How many cubes have no face painted ?
(a) 0 (6) 4 (e) 8 id) 12
- (2) How many cubes have only one face painted ?
(a) 8 (b) 16 (c) 24 (d) 28
- (3) How many cubes have only two faces painted ?
(a) 8 (6) 16 (c) 20 (d) 24
- (4) How many cubes have three faces painted ?
(a) 0 (6) 4 (c) 6 (d) 8
- (5) How many cubes have three faces painted with different colours ?
(a) 0 (6) 4 (c) 8 (d) 12
- (6) How many cubes have two faces painted red and black and all other faces unpainted ?

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- (a) 4 (6) 8 (c) 16 <d> 24

(5) How many cubes have one face green and one of the adjacent faces black or red ?

- (a) 8 (6) 16 (c) 24 id) 28

Q.7. Directions : The six faces of a cube are coloured black, brown, green, red, white and blue, such that

- (i) Red is opposite black
- (ii) Green is between red and black
- (iii) Blue is adjacent to white
- (iv) Brown is adjacent to blue
- (v) Red is at the bottom.

Answer the following questions based on this information

(U.D.C. 1995)

(1) Which colour is opposite brown ?

- (a) White (6) Red (c) Green (d) Blue

(2) The four adjacent colours are

- (a) Black, Blue, Brown, Red (6) Black, Blue, Brown, White
<c> Black, Blue, Red, White id) Black, Brown, Red, White

(3) Which of the following can be deduced from (i) and (v) ?

- (a) Black is on the top (6) Blue is on the top
(c) Brown is on the top (d) Brown is opposite Black

Q.8. Directions : A cube is painted blue on all faces is cut into 125 cubes of equal size. Now, answer the following questions : (M.B.A. 1994)

(1) How many cubes are not painted on any face ?

- (a) 8 (6) 16 (c) 18 (d) 27 <c> 54

(2) How many cubes are painted on one face only ?

- (a) 8 (6) 16 (c) 36 (d) 54 (e) None of these

Q.9. Directions : A cube is coloured orange on one face, pink on the opposite face, brown on one face and silver on a face adjacent to the brown face. The other two faces are left uncoloured. It is then cut into 125 smaller cubes of equal size. Now, answer the following questions based on the above statements :

(1) How many cubes have at least one face coloured pink ?

- (a) 1 (6) 9 (c) 16 id) 25

(2) How many cubes have all the faces uncoloured ?

- (a) 24 ib) 36 (c) 48 (</) 64

<3> How many cubes have at least two faces coloured ?

- ia) 19 (b) 20 (c) 21 Id) 23

(4) How many cubes are coloured orange on one face and have the remaining faces uncoloured ?

- (a) 8 (6) 12 (c) 14 (d) 16

(5) How many cubes are coloured silver on one face, orange or pink on another face and have four uncoloured faces ?

- (a) 8 (6) 10 (c) 12 (</) 16

Q.10. Directions : The length of each side of a cube is 5 cms. The outer border of the width of 1 cm is painted yellow on each side and the remaining space enclosed by this 1 cm. path, is painted pink. This cube is cut into 125 smaller cubes of each side 1 cm. When these smaller cubes are separated:

(1) How many cubes have all the faces uncoloured ?

- (a) 0 (b) 9 <c> 18 id) 27

(2) How many cubes have three faces coloured yellow ?

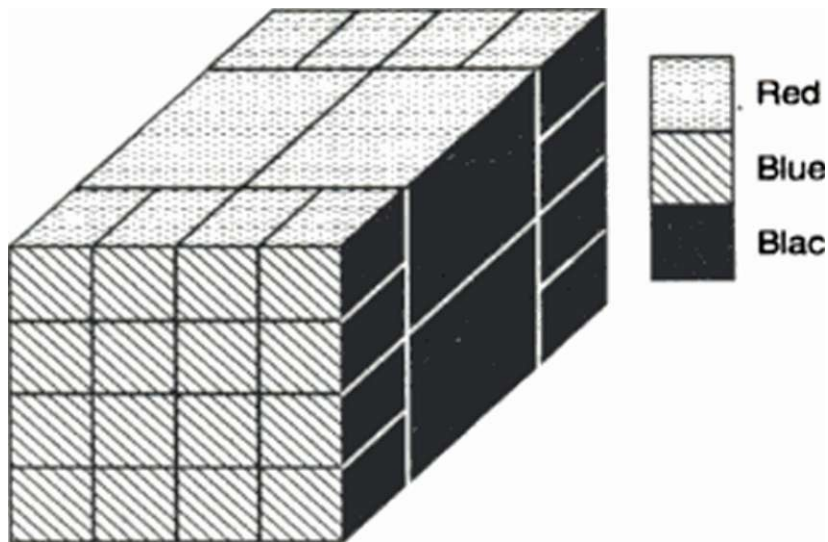
- (a) 2 (6) 4 (c) 8 id) 10

<3> How many cubes have at least two faces coloured yellow ?

- (a) 24 id> 44 tc) 48 id) 96

- (4) How many cubes have one face pink and an adjacent face yellow ?
 (a) 0 (6) 1 (c) 2 (d) 4
- (5) How many cubes have at least one face coloured ?
 (a) 27 (6) 48 (c) 98 (d) 121

Q.11. Directions : A solid cube has been painted yellow, blue and black on pairs of opposite faces. The cube is then cut into 36 smaller cubes such that 32 cubes are of the same size while 4 others are of bigger size. Also no face of any of the bigger cubes is painted blue.



- (1) How many cubes have at least one face painted blue ?
 (a) 0 (6) 8 (c) 16 (d) 32
- (2) How many cubes have only one face painted ?
 (a) 0 (b) 4 (c) 8 (d) 12
- (3) How many cubes have only two faces painted ?
 (a) 24 (b) 20 (c) 16 (d) 8
- (4) How many cubes have two or more faces painted ?
 (a) 36 (6) 34 (c) 28 (d) 24
- (5) How many cubes have only three faces painted ?
 (a) 8 (6) 4 (c) 2 (d) 0
- (6) How many cubes do not have any of their faces painted yellow ?
 (a) 0 (6) 4 (c) 8 (d) 16
- (7) How many cubes have at least one of their faces painted black ?
 (a) 0 (6) 8 (c) 16 (d) 20
- (8) How many cubes have at least one of their faces painted yellow or blue ?
 (a) 36 (6) 32 (c) 16 (d) 0
- (9) How many cubes have no face painted ?
 (a) 8 (6) 4 (c) 1 (d) 0
- (10) How many cubes have two faces painted yellow and black respectively ?
 (a) 0 (b) 8 (c) 12 (d) 16

Q.12. Directions : A cube is coloured Red on two opposite faces, Blue on two adjacent faces and Yellow on the two remaining faces. It is then cut into two halves along the plane parallel to the Red faces. One-piece is then cut into four equal cubes and the other one into 32 equal cubes. Now answer the following questions based on the above statement: (Hotel Management, 1995)

- (1) How many cubes do not have any coloured face ?
 (a) 0 (6) 2 (c) 4 (d) 8
- (2) How many cubes do not have any Red face ?

- (a) 8

ib) 16

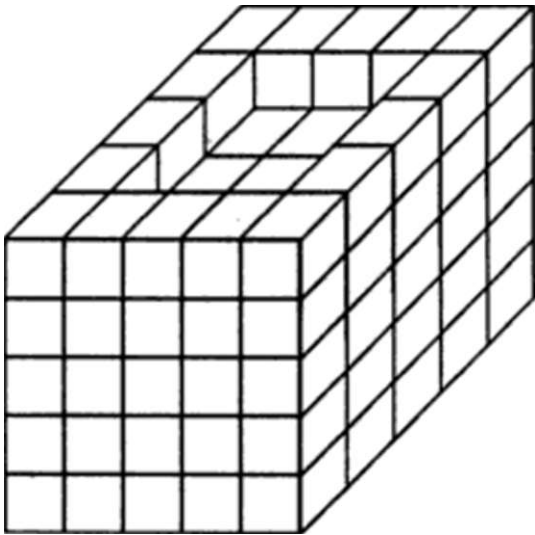
ic) 20

id) 24
- (3) How many cubes have at least two coloured faces ?
ia) 20 (6) 24 (c) 28 id) 32
- (4) Haw many cubes have each a Yellow face with other faces blank ?
ia) 4 (b) 14 (c) 16 (d) 17
- (5) How many cubes have at least one Blue face ?
ia) 4 ib) 14 (c) 16 (d) 20
- Q.13. Directions :** A cube is painted red on two tidjacent faces and on one opposite face, yellow on two opposite faces and green on the remaining face. It is then cut into 64 equal cubes.

How many cubes have only one red coloured face ?

(S.S.C. 1993)

ia) 4 (6) 8 ' ic) 12 (d) 16



- Q.14. Directions :** Some equal cubes are arranged in the form of a solid block as shown in the adjoining figure. All the visible surfaces ofthe block (except the bottom) are then painted.

(1) How many cubes do not have any of the faces painted ?
(cr) 27 ib) 32 (c) 36 id) 40

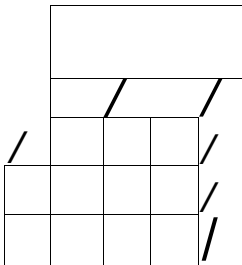
(2) How many cubes have only one face painted ?
ia) 9 ib) 45 (c) 57 id) 62

(3) How many cubes have only two faces painted ?
(a) 0 (b) 16 (c) 20 id) 24

(4) How many cubes have only three faces painted ?
(a) 4 ib) 12 (c) 16 id) 20
- Q.15. Directions :** A solid black is formed by arranging some cubes of equal dimensions, as shown in the adjoining figure. Whole of the block except the base has been coloured pink. Answer the questions based upon this information:

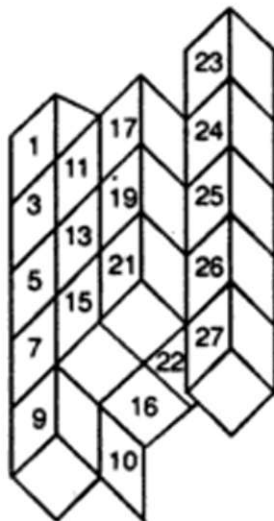
2 :

- (1) How many cubes have four faces coloured ?
(a) 0 (6) 1 (c) 2 (rf) 3
- (2) How many cubes have exactly three faces coloured ?
(a) 2 (6) 4 (c) 8 (d) 11
- (3) How many cubes have exactly two faces colottr^T0
ia) 9 (6) 11 icT 13 id) 15
- (4) How niany cubes have only one face coloured ?
id) 22 (6) 24 " ic) 25



id) 27

- (5) How many cubes are not coloured on any face ?
 (a) 6 (b) 8 (c) 9 (rf) 11
- Q.16.** The minimum number of colours required to paint all the sides of a cube that no two adjacent faces may have the same colours, is : **(M.BA. 1094)**
 (a) 1 (b) 2 (c) 3 (rf) 4
- Q.17. Directions:** Three adjacent faces of a cube are coloured blue. The cube is then cut (once horizontally and once vertically to form four cuboids of equal size each of these cuboids is coloured pink on all the uncoloured faces and is then cut (as before) into four cuboids of equal size.
- (1) How many cuboids have two faces coloured pink ?
 (a) 1 (b) 3 (c) 4 (id) 6
- (2) How many cuboids have three faces coloured pink ?
 (a) 9 (b) 7 (c) 5 (rf) 3
- (3) How many cuboids have three faces coloured blue ?
 (a) 4 (b) 2 (c) 1 (d) 0
- Q.18. Directions :** In the adjoining figure there are 27 cubes numbered from 1 to 27. Answer the questions that follow :
- (1) Number of cubes which have maximum number of faces touching the other cubes is :
 (a) 1 (b) 2 (c) 3 (rf) 4
- (2) Number of cubes which have minimum number of faces touching the other cubes is :
 (a) 1 (b) 2 (c) 4 (rf) 6
- (3) Which of the cubes 3, 8, 20 or 27 touches the maximum number of cubes ?
 (a) 3 (b) 8 (c) 20 (d) 27
- (4) Which of the cubes 6, 13, 19 or 25 touches maximum number of cubes ?
(a) 6 (b) 13 (c) 19 (d) 25

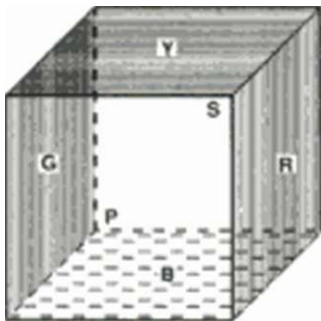


- (5) Which of the cubes 7, 21, 22, or 23 touches minimum number of cubes ?**
(a) 7 (b) 21 (c) 22 (d) 23
- (6) What cube is hidden under cube 19 ?
 (a) 18 (b) 20 (c) 24 (rf) 25
- (7) What cube is hidden under cube 13 ?
 (a) 5 (b) 14 (c) 15 (rf) 19
- (8) How many cubes are hidden and so could not be numbered ?
 (a) 8 (b) 6 (c) 5 (rf) 4
- (9)** How many cubes have their faces touching four other cubes ?
 (a) 9 (b) 10 (c) 12 (rf) 13
- (10)** The cubes having their five faces touching other cubes are :
 (a) 7, 14, 20 (b) 13, 14, 20 (c) 13, 18, 22 (rf) 13, 14, 18, 20, 22

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- 3. (d): There are 8 cubes in layer I, 4 cubes in layer II, 4 cubes in layer III and 8 cubes in layer IV which have only two faces painted. Thus, there are $8 + 4 + 4 + 8 = 24$ such cubes.
- 4. (</): Four corner cubes in layer I and four corner cubes in layer IV have three faces painted. Thus, there are 8 such cubes.
- 5. (c): Four corner cubes in layer I and four corner cubes in layer IV have three faces painted with different colours. Thus, there are 8 such cubes
- 6. (6): There are four cubes in layer I and four cubes in layer IV which have two faces painted red and black and all other faces unpainted. Thus, there are 8 such cubes
- 7. (6): There are four cubes in layer I and four cubes in layer IV which have only one face painted red and all other faces unpainted. Thus, there are 8 such cubes.
- 8. (d): There can be no cube which has two of its faces both painted with the same colour
- 9. (a): There are $4 + 4 = 8$ cubes in layer I and $4 + 4 = 8$ cubes in layer IV which have one face painted blue and one face painted red. Thus, there are 16 such cubes.
- 10. (a): There are four layers of 16 cubes each. Thus, there are $16 \times 4 = 64$ cubes in all.

Sol. 4: On the basis of the given details, the cube will be painted as indicated in the following figure.



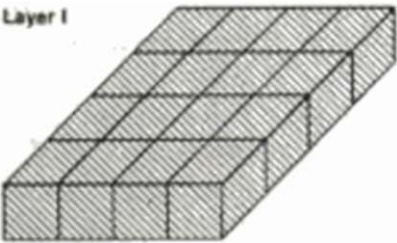
Here Y stands for Yellow ; 'R' for Red ; 'B' for Brown ; 'G' for Green ; V for Pink and 'S' for Silver. The colour of each face is indicated at the centre of each face.

- 1. (6): The face opposite to Red is Green.
 - 2. (c): The upper face is painted yellow.
 - 3. (d): Clearly, the faces adjacent to Green are Pink, Silver, Yellow and Brown.
 - 4. (a): Clearly, the face opposite to silver is Pink.
 - 5. (6): The faces adjacent to Red face are Silver, Pink, Brown and Yellow
- Sol. 5:** The given figure shows the cube coloured red on all faces, and divided into 64 smaller cubes :

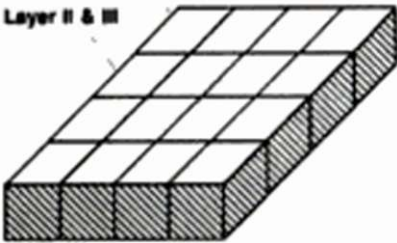


The figure may be analysed by dividing it into four horizontal layers :

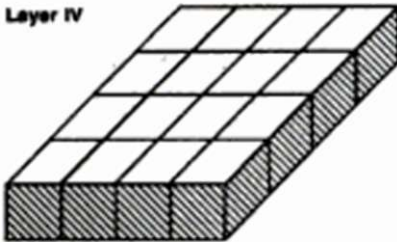
In **layer I;** the four central cubes have only one face coloured four cubes at the corner have three faces coloured and the remaining 8 cubes have two faces coloured.



In **each of the layers II & ID,** the four central cubes have no face coloured, the four cubes at the corner have two faces coloured and the remaining 8 cubes have only one face coloured.



In **layer IV,** the four central cubes have only one face coloured, four cubes at the corner have three faces coloured and the remaining 8 cubes have two faces coloured.



- 1.(c): Four central cubes in layer II and four central cubes in layer III have no face coloured. Thus there are 8 such cubes.
- 2. (d): Four central cubes in layer I, 8 cubes in layer II, 8 cubes in layer III and four central cubes in layer IV have only one face coloured. Thus, there are 4 + 8 + 8 + 4 = 24 such cubes.
- 3. (a): None of the cubes can have its opposite faces coloured.
- 4. (c): Four corner cubes in layer I and four corner cubes in layer IV have three faces coloured. Thus, there are 8 such cubes.

Sol. 6 : The figure analysis is the same as in the solution of Q. 3.

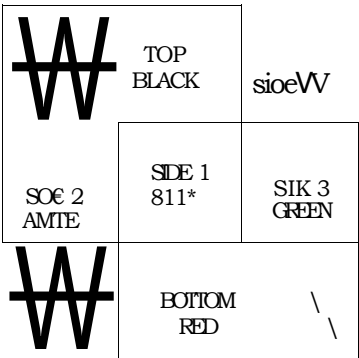
- 1. (c): There are 8 cubes having no face painted.
- 2. (c): There are 24 cubes having only one face painted.
- 3. (d): There are 24 cubes having only one face painted and 24 cubes having only two faces painted.

Thus, there are 24 + 24 = 48 cubes having less than three faces painted.

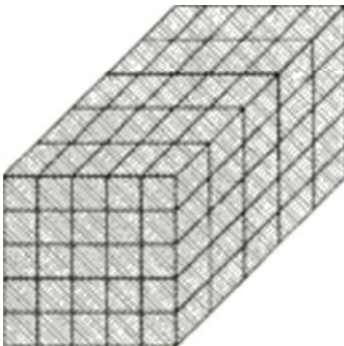
- 4, (6): There are 8 cubes having three faces painted.
- 5. (c): In Q. 3. substitute blue colour by green colour. 8 cubes in layer I, 4 cubes in layer II, 4 cubes in layer III and 8 cubes in layer IV have one face green and one of the adjacent faces black or red. Thus there are 8 + 4 + 4 + 8 = 24 such cubes.

Sol. 7 : On the basis of the given details, the cube will be coloured as indicated in the figure shown.

- 1. (a): As is clear from the figure side 4 is coloured brown and opposite to it lies the side 2 which is coloured white. Therefore, white colour is opposite brown.
- 2. (d): Black, Brown, Red, White are adjacent colours which lie respectively on top, side 4, bottom and side 2.
- 3. (a) : Red is opposite Black, and Red is at the bottom implies 'Black is on the top'.

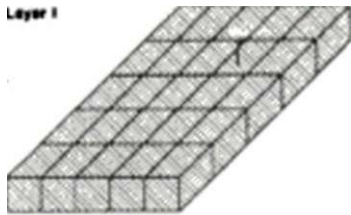


Sol. 8.: The following figure shows the cubes painted blue on all face and divided into 125 smaller cubes:

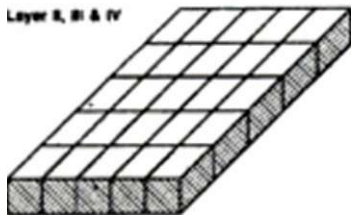


The figure may be analysed by dividing it into five horizontal layers :

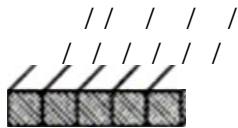
In layer I; the nine central cubes have only one face painted, four cubes at the corner have three faces painted and the remaining 12 cubes have two faces painted.



In each of the layer* II, III & IV; the nine central cubes have **no** face painted, the four cubes at the corner have two faces painted and the remaining 12 cubes have **one** face painted .

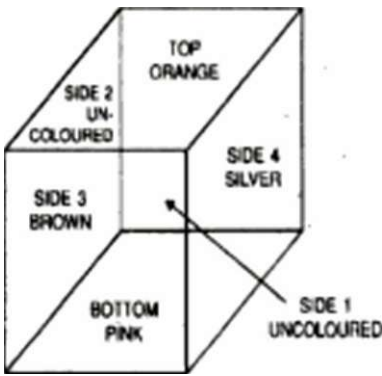


In layer V; the 9 central cubes have only one face painted, four cubes at the corner have three faces painted and the remaining 12 cubes have two faces painted.

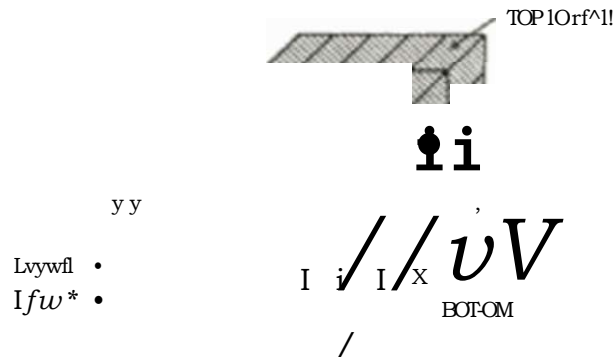


- 1. *id*): There are 9 central cubes in each of the layers II, III and IV which have no face painted. Thus, there are $9 \times 3 = 27$ such cubes.
- 2. (d): There are 9 cubes in layer I, 12 cubes in each of the layers II, III and IV, 9 cubes in layer V which are painted on one face only. Thus, there are $9 + 12 \times 3 + 9 = 54$ such cubes.

Sol. 9: On the basis of the given details, the cube will be coloured as indicated in the adjoining figure:



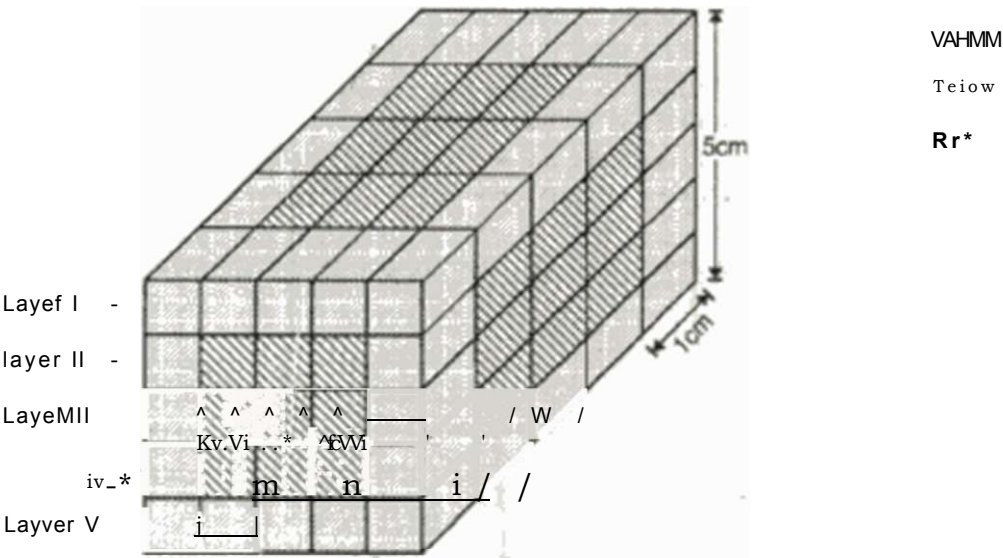
This cube is divided into 125 smaller cubes as shown in the figure below.



The figure may be analysed by assuming it to be made up of five horizontal layers :

- 1. (d): All the 25 cubes in layer V have at least one face coloured pink.
- 2. (c): In each of the layers II, III and IV, all the cubes except those which lie along the sides 3 and 4, have all the faces uncoloured. Thus, there are 16 such cubes in each of these three layers i.e. there are $16 \times 3 = 48$ such cubes in all.
- 3. (c): There are 8 cubes in layer I, 1 cube each in the layers II, III and IV and 8 cubes in layer V which have two faces coloured. Also, there is 1 cube in layer I, and 1 cube in layer V which have three faces coloured. Thus, there are $8 + 1 \times 3 + 8 + 1 + 1 = 21$ cubes having at least two faces coloured.
- 4. (d): The cubes coloured orange on one face and having the remaining faces uncoloured, lie in layer I except along the sides 3 and 4. Thus, there are 16 such cubes.
- 5. (a): There are 4 cubes in layer I along side 4 which are coloured silver on one face, orange on another face and having four uncoloured faces. Also, there, there 4 cubes in layer V along side 4 which are coloured silver on one face, pink on another face and having four uncoloured faces. Thus, there are 8 cube* of required type.

Sol. 10 2 The following figure shows the cube which is painted as stated in the question, and then divided into 125 smaller cubes :



The figure may be analysed by assuming the larger cube to be made up of five horizontal layers.

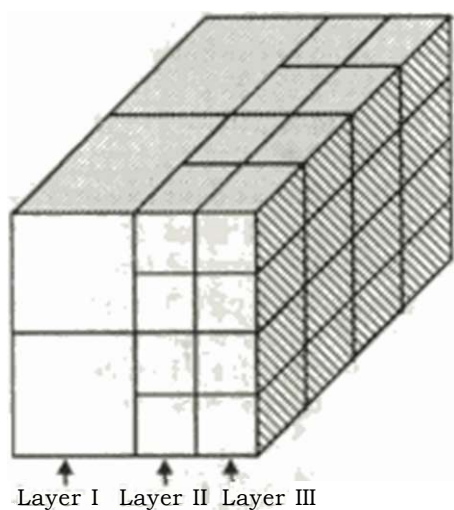
- 1.(rf): 9 cubes in the centre of each of the layers II, III and IV have all the faces uncoloured. Thiu, there are $9 \times 3 = 27$ such cubes.
- 2. (c): 4 corner cubes ir. layer I and 4 corner cubes in layer V have three faces coloured yellow. Thus, there are 8 such cubes.

3. (6): 12 cubes in layer I. 4 cubes in each of the layers II, III and IV and 12 cubes in layer V have two faces coloured yellow. 4 cubes in layer I and 4 cubes in layer V have three faces coloured yellow. Thus, there are $12 + 4 \times 3 + 12 + 4 + 4 = 44$ cubes having at least two faces coloured yellow.
4. (a): There is no cube having one face pink and an adjacent face yellow.
5. (c): There are 125 cubes in all and out of these 27 cubes have all faces uncoloured. Thus, there are $125 - 27 = 98$ cubes having at least one face coloured.
- Sol. 11 : In the figure there are 36 cubes, 32 of which are of the same size and 4 others are of bigger size. Clearly, each side of bigger cubes is twice as large as that of smaller cubes. Also, since no face of any of the larger cubes is painted blue, so, each one of the larger cubes has one face painted red, one face painted black and all other faces unpainted.
1. (c): Cubes having at least one face painted blue are those which lie along the two surfaces painted blue. Since each blue surface has 16 cubes, so, there are $16 \times 2 = 32$ such cubes.
2. (c): The cubes having only one face painted are the 4 central cubes along each of the two surfaces painted blue. Thus, there are $4 \times 2 = 8$ such cubes.
3. (6): Leaving the 4 central cubes on each of the blue surface and 8 cubes at the edges of the block; all the rest have two faces painted. Thus, the cubes having only two faces painted are 8 cubes on each of the blue surfaces and 4 larger cubes i.e. there are $(8 \times 2) + 4 = 20$ such cubes.
4. (c): There are no cubes having more than three faces painted. 8 cubes at the edges of the block have three faces painted and as calculated above, 20 cubes have two faces painted. Thus, $8 + 20 = 28$ cubes have two or more faces painted.
5. (a): The 8 cubes at the edges of the block have three faces painted.
6. (d): The cubes having at least one face painted yellow are the 10 cubes along the top surface and 10 cubes along the bottom surface i.e. 20 cubes having at least one face painted yellow.
Thus, the number of cubes having none of their faces painted yellow are $36 - 20 = 16$.
7. (d): The cubes having at least one of the faces painted black are the 10 cubes along each of the black surfaces, i.e. there are 20 such cubes.
8. (a): The number of cubes having at least one of the faces painted yellow is 20. Also, the number of cubes having at least one of the faces painted blue is 32 (Those lying along the two blue surfaces). But there are 16 cubes lying along the 4 edges common to blue and yellow surfaces.
Thus, the cubes having at least one of their faces painted yellow or blue is $(20 + 32 - 16) = 36$.
9. (d): The number of cubes having at least one of the faces painted are the 16 cubes, each along the surfaces painted blue and 4 larger cubes. Thus there are $(16 \times 2) + 4 = 36$ cubes having at least one of the faces painted. Since, there are 36 cubes in all, therefore, number of cubes having no face painted = $36 - 36 = 0$.
10. (c): Clearly, there are two small and one larger cubes i.e. 3 cubes along each of the edges common to yellow and black surfaces, which have one face painted yellow and one face painted black. Thus, there are $(4 \times 3) = 12$ such cubes.

Sol. 12 : The following figure shows the cube which is coloured and cut as stated in the question :

The figure may be analysed by assuming the larger cube to be made up of three vertical layers.

1. (c): Four central cubes in layer II do not have any coloured face.
2. (6): The 16 cubes in layer II do not have any red face.
3. (a): All the four cubes in layer I have three coloured faces, four cubes in the corner of layer II have two coloured faces and except for the four central cubes all the



Yellow

Blue

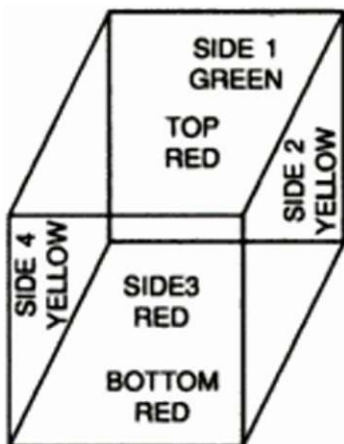
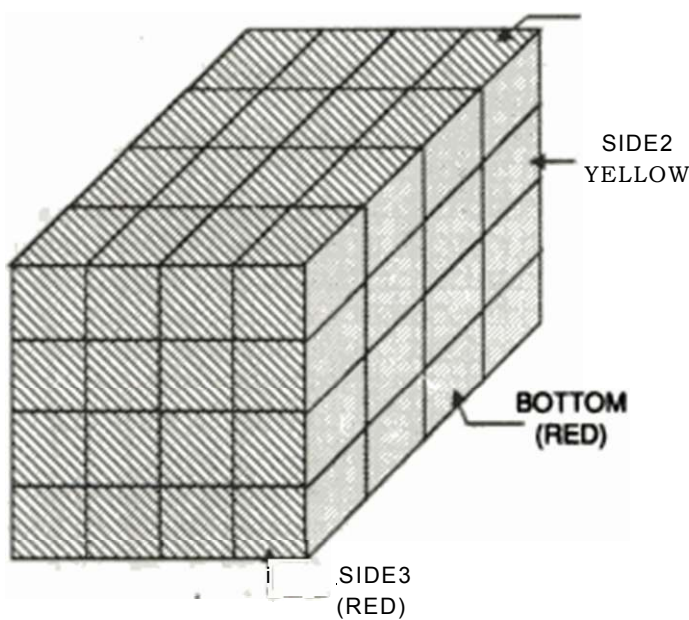
remaining 12 cubes in layer III have two or three faces coloured. Thus, there are $4 + 4 + 12 = 20$ cubes having at least two coloured faces.

4. (a): There are four cubes each having one yellow face and all other faces blank, these cubes lie in layer II.

5. (d): All the four cubes in layer I and 8 cubes in each of the layers II and III have at least one blue face. Thus, there are $4 + 8 + 8 = 20$ such cubes

Sol 13 (c): Fig. 1 shows the cube which is painted as stated in the question.

When this cube is divided into 64 smaller cubes, we get the figure as shown in Fig. 2.

**Fig 1****Fig 2**

Clearly the cubes which have only one face red coloured and all other faces uncoloured are the four central cubes at each of the three faces of the larger cube the top, side 3 and the bottom. Thus, there are $4 \times 3 = 12$ such cubes.

Sol. 14: In the figure there are 16 columns of 5 cubes each and 9 columns of 4 cubes each i.e. a total to 116 cubes.

1. (a): The cube® having at least one of their faces painted are the 16 columns of 5 cubes each placed along the periphery and 9 cubes of the top.

Thus, number of cubes having at least one face painted = $(16 \times 5) + 9 = 89$.

Hence, the number of cubes having no face painted = $116 - 89 = 27$.

- 2 Ac)** The cubes having only one face painted are the 12 central cubes on each side of the block and 9 cubes of the top.

InNote here that the bottom surfacc of the block is not painted)

Thus. $(4 \times 12) + 9 = 57$ cubes have only one face painted

3. (6): The cubes having only two faces painted are four cubes along each of the vertical edges leaving the top one in each of these edges. Thus, there are $(4 \times 4) = 16$ such cubes.
4. (c): The cubes placed on the top of all the 16 coloumns forming the boundary of the block are the onces which have exactly three faces painted. Thus, there are 16 cubes having three faces painted.

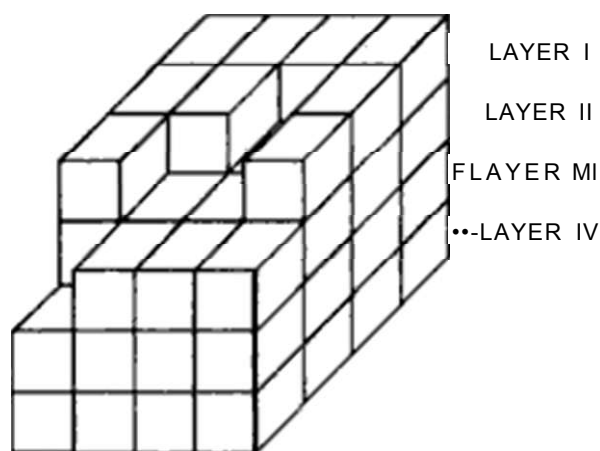
Sol. 15.: The given figure may be analysed by assuming the block to be made up of four horizontal layers, as shown in the adjoining figure :

In layer I; there are 2 cubes having four faces coloured, 5 cubes having three faces coloured. 2 cubes having two faces coloured.

In layer II; there are 2 cubes having three faces coloured, 4 cubes having two faces coloured. 8 cubes having one face coloured and 1 cube having no face coloured.

In layer III: there is 1 cube having three faces coloured, 3 cubes having two faces coloured 8 cubes having one face coloured and 4 cubes having no face coloured.

In layer IV; there are 4 cubes having two faces coloured. 8 cubes having one face coloured and 4 cubes having no face coloured.



- 1.(c). There are 2 cubes (in layer I) having four faces coloured.
2. (c): There are 5 cubes in layer I, ? cubes in layer II & 1 cube in layer III. having three faces coloured Thus, there are $5 + 2 + 1 = 8$ such cubes
3. ic). There are 2 cubes in layer I, 2 cubes in layer II, 3 cubes in layer III and 4 cubes in layer IV having two faces coloured Thus, there are $2 + 4 + 3 + 4 = 13$.
4. (6): There are 8 cubes in layer II, 8 cubes in layer III and 8 cubes in layer IV havmg only one face coloured. Thus, there are $8 + 8 + 8 = 24$ such cubes
5. (c): There is 1 cube in layer II, 4 cubes in layer III and 4 cubes in layer IV having no face coloured. Thus, there are $1 + 4 + 4 = 9$ such cubes.

Sol. 16 (c): Opposite faces can have the same colour and there are six faces in a cube

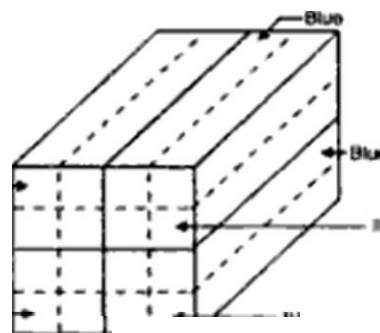
Sol. 17.: The adjoining figure shows the cube coloured and cut into four cuboids as stated in the question :

These four cuboids are separated and coloured pink on all uncoloured faces. Each of these four cuboids is then cut into four other cuboids as shown by the dotted lines in the figure Thus, we get 4 sets of 4 cuboids each.

In set I & IV : 2 cuboids have 2 faces blue. 2 faces pink and 2 faces uncoloured each.
2 cuboids have 1 face blue, 3 faces pink and 2 faces uncoloured each

In set II : 2 cuboids have 2 faces blue, 2 faces pink and 2 faces uncoloured each

1 cuboid has 3 faces blue, 1 face pink and 2 faces uncoloured each.



1 cuboid has 1 face blue, 3 faces pink and 2 faces uncoloured each.

In set HI: All the four cuboids have 1 face blue, 3 faces pink and 2 faces uncoloured each.

1. (d): There are 2 cuboids in set I, 2 cuboids in set II and 2 cuboids in set III having 2 faces pink in each. Thus, there are $2 + 2 + 2 = 6$ such cubes
 2. (a): There are 2 cuboids in set I, 1 cuboid in set II, 4 cuboids in set III and 2 cuboids in set IV having 3 faces pink each. Thus, there are 9 such cuboids.
 3. (c): There is only one cuboid having three faces blue. This cuboid lies in set II.
- SoL 18:** The number of each of the cubes is given along with the number of its faces touching other cubes :—
- I. $\rightarrow 2$; 2. $\rightarrow 2$; 3. $\rightarrow 4$; 4. $\rightarrow 4$; 6. $\rightarrow 4$; 6. $\rightarrow 4$; 7. $\rightarrow 4$; 8. $\rightarrow 4$; 9. $\rightarrow 2$; 10. $\rightarrow 2$;
 II. $\rightarrow 4$; 12. $\rightarrow 4$; 13. $\rightarrow 5$; 14. $\rightarrow 6$; 15. $\rightarrow 4$; 16. $\rightarrow 4$; 17. $\rightarrow 3$; 18. $\rightarrow 4$; 19. $\rightarrow 4$;
 20. $\rightarrow 4$; 21. $\rightarrow 3$; 22. $\rightarrow 4$; 23. $\rightarrow 1$; 24. $\rightarrow 3$; 25. $\rightarrow 3$; 26. $\rightarrow 3$; 27. $\rightarrow 1$.
1. (a): The cubes having the maximum number of faces touching other cubes are those touching 5 other cubes. These are 13, 14, and 20. i.e. there are 3 such cubes.
 2. (6): The cubes having the minimum number of faces touching other cubes are those touching 1 cube only. These are 23 and 27. i.e. there are 2 such cubes
 3. (c): Only the cube 20 touches 6 other cubes i.e. maximum number of cubes.
 4. (6): Only the cube 13 touches 5 other cubes i.e. maximum number of cubes.
 6. (d): Only the cube 23 touches 1 other cube i.e. minimum number of cubes.
 6. (b): The number of the cube below any cube is one greater than that of the upper cube
 Cube 20 lies below cube 19.
 7. (b): The cube hidden below cube 13. is cube 14.
 8. (a): The cubes hidden are 2, 4, 6, 8, 12, 14, 18 and 20 which cannot be numbered since none of their faces is visible.
 9. (d): The cubes having their faces touching four other cubes are 3, 4, 5, 6, 7, 8, 11, 12, 5, 16, 18, 19 and 22 Thus, there are 13 such cubes.
 10. (6): Cubes 13, 14 and 20 have their faces touching five other cubes.

QUANTITATIVE APTITUDE

-Dr. R.S. Aggarwal

**For Bank P.O., S.B.I.P.O., M.B.fl., N.D.R., C.D.S.,
 Hotel Management, I. Tax & Central Excise, C.B.I.,
 Railways, U.C.fl.fl.O., G.I.C.R.R.O., Rstt. Grade.
 U.D.C., etc.**

**With whole lot of questions, fully solved by short - cut
 methods.**

CONSTRUCTION OF BOXES

When a cube or a cuboid is unfolded, it may appear in any of the following forms :

Form 1 :

	1	
h	3	'I
	5	
	6	

In this case :

1 lies opposite 5;

2 lies opposite 4;

3 lies opposite 6.

Form 2 :

In this case

1 lies opposite 6;

2 lies opposite 4;

3 lies opposite 5.

Form 3 :

		^	
2	3		
	4		
	5	6	

In this case

1 lies opposite 4;

2 lies opposite 6;

3 lies opposite 5.

Form 4 :

1	
2	
3	4
	5
	6

In this case

1 lies opposite 3;

2 lies opposite 5;

4 lies opposite 6.

Form 5 =

	1	
I'	3	
	5	

In this case

will be one of the faces of the cube,

which lies opposite 3;

2 lies opposite 4;

1 lies opposite 5.

Form 5

	E	
	1	
H ₂	3	
	5	
	- r	

In this case

m will be one of the faces of the cube,

which lies opposite 3;

2 lies opposite 4;

1 lies opposite 5.

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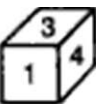
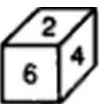
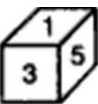
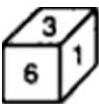
EXERCISE 14C

Directions : The figure (X) given on the left hand aide, In each problem, la folded to form a box. Choose from amongst the alternatives (a), (b), (c) and (d), the boxes that are similar to the box formed.

1.

	2	
5	1	
	4	
	6	3

(X)



(a) (b) (c) (d)

(X)

(a) (*) (c) (d)

(X)



(a) (b) (c) (d)

(X)



(a) (b) (c) (d)

5.

(X)

(Asstt. Grade, 1992)

6.

(X)

- (A) (B) (C) (D)
- (A) (B) (C) (D)

- (a) A and B only (to) B and C only (a) A and C only (to) B, C and D only
- (c) B and D only (cf) A, B, C and D (c) B and D only (d) C and D only

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ANSWERS

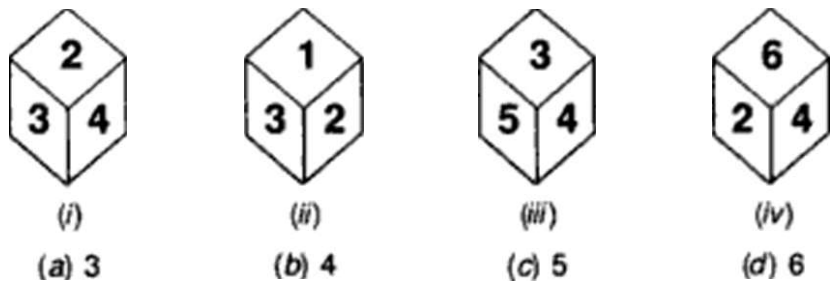
1. (d): Fig (X) is the same as form 3. so, when cube is formed, 2 lies opposite 4; 1 lies opposite 6 and 5 lies opposite 3. Hence, the pairs 2 & 4, 1 & 6 & 5 & 3 cannot occur at adjacent faces. So, only cube (d) can be formed.
2. (a): When fig. (X) is folded to form a cube, the two rectangular portions combine to form the top of the cube. So, cube (a) will be formed.
3. (6) Fig (X) is the same as form 3. So, when it is folded to form a cube, the half shaded face will lie opposite the face bearing the rhombus. So, the possibility of cubes (a) and (c) is ruled out. Out of the cubes (6) and (c), the pattern in figure (x) shows that cube (6) will be formed.
4. (cf): Fig. (x) is the same as form 5. The four triangular portions will combine to form a face of the type $I\wedge T$ which lies opposite to the face bearing the circle. So, none of the cubes (o), (6) or (c) can be formed.
5. (cf): Fig. (x) is the same as form 2. So, the half shaded faces lie opposite to each other. The pattern in fig. (x) shows that on folding, each of the cubes A, B, C and D are possible.
6. (6): Fig. (x) is the same as form 1. So, the half shaded face and the completely shaded faces lie opposite to each other. Therefore, the cube (A) cannot be formed. Moreover, the pattern in fig (x) shows that out of the cubes B, C and D, only the cubes B and D can be formed by folding fig. (x).
7. (d): Fig (x) is the same as form 4- So, the faces bearing the circle and the square lie opposite to, each other. Therefore, the cubes (A) and (B) cannot be formed. The pattern in fig. (x) Shows that both the cubes (O and (D) can be formed by folding fig. (x).
8. (d): Fig. (x) is the same as form 5. The pattern on fig. (x) shows that all the cubes (A), (B), (C) and (D) can be formed by folding fig. (x).
9. (b): Fig. (x) is the same as form 3. So, when the cube is formed, F lies opposite B; E lies opposite C; D lies opposite A. Hence, the pairs F & B, E & C and D & A cannot occur at adjacent faces. So, only cube (B) can be formed.
10. (a): The pattern on fig. (x) shows that the cubes (A) and (D) can be formed by folding fig. (x).
11. (d): The pattern on fig. (x) shows that the cubes (B), (C) and (D) can be formed by folding **fig. (x).**
12. (b): When fig. (x) is folded to form a cube with one of the faces as le|«| Therefore, cube, (B) can be formed by folding fig. (x) None of the cubes A, C or D can be formed by folding fig. (x).
13. (a): Fig. (x) is similar to form 1. So, the two rectangular shaded portions form two faces of the cuboid. Therefore, the cuboids (B) and (D) cannot be obtained by folding fig. (x). Both the cuboids (A) and (C) can be obtained by folding fig. (x)
14. (d): The pattern in fig. (x) shows that each one of the cubes (A), (B), (C) and (D) can be formed by folding fig. (x).
15. (c): The shaded face and the face bearing the square will lie opposite to each other So, the cubes (B) and (D) cannot be formed by folding fig. (x). The pattern shows that both the cubes (A) and (C) can be formed by folding fig. (x).
16. (b): One of the black faces lies opposite the face bearing the dot. So, cube (B) cannot be formed by folding fig. (x). The pattern in fig. (x) shows that each of the cubes (A), (C) and (D) can be formed by folding fig. (x).
17. (a): The faces bearing the dot and the shading lie opposite to each other. So, the cubes (B) and (D) cannot be formed by folding fig. (x). Two of the blank faces lie opposite to each other. So, the three blank faces cannot lie adjacent to each other. Therefore, the cube (C) also cannot be formed. The pattern shows that cube (A) can be formed by folding fig. (x).
18. (b): The pattern in fig. (x) shows that when fig. (x) is folded, only the cubes (A) and (C) can be formed.

19. (6): The pattern in fig. 00 shows that except cube (C) each other cube can be formed by folding fig. (x).
20. (c): When fig. (x) is folded to form a cube, the two rectangular portions combine to form a face . This face lies opposite the face bearing the square. Also, two of the blank faces lie opposite to each other. So the three blank faces cannot lie adjacent to each other. Hence, cube (B) cannot be formed. The pattern in fig. (x) shows that all other cubes can be formed by folding fig. (x).
21. (d): The pattern on fig. (x) shows that only cube (d) can be formed by folding fig. (x).
22. (a): The fig. (x) is similar to form 2. So, the two half shaded portions lie opposite to each other. Therefore, the cubes (6) and (c) cannot be formed by folding fig. (x). The pattern in fig. (x) shows that cube (d) cannot be formed and only cube (a) can be formed by folding fig. (x).
23. (d): The two half shaded faces lie opposite to each other when fig. (x) is folded. So, the cube (B) cannot be formed. The pattern in fig (x) shows that, out of the remaining three cubes, only the cubes (A) and (C) can be formed by folding fig. (x).
24. (a): When fig. (x) is folded to form a cube, the two half shaded faces lie opposite to each other. So, the cube (D) cannot be formed. The pattern in fig. (x) shows that out of the remaining three cubes, only the cubes (A) and (C) can be formed by folding fig. (x).
25. (c): Fig. (x) is similar to form 5. So, the four triangular portions combine to form a single face of the cube when fig. (x) is folded. Therefore, the cubes (A) and (D) will not be formed. The pattern on fig. (x) shows that both the cubes (B) and (C) can be formed by folding fig. (x).
26. (a): The pattern on fig. (x) and also the fact that the faces are rectangle, indicate that only fig. (A) can be obtained by folding fig. (x).
27. (d): The pattern in cube (x) shows that only fig. (d) can be formed by unfolding the cube (x).
28. (c): The given figure is similar to form 3. So, three dots would lie opposite the face bearing five dots.
29. (d): The given figure is similar to form 4. So, six dots would lie opposite the face having three dots.

PROBLEMS ON DICE

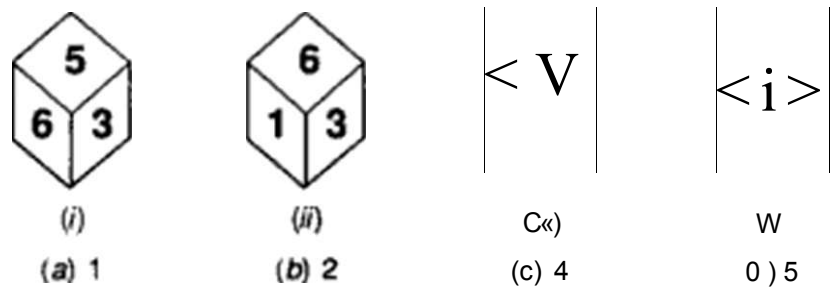
Sometimes we are given figures showing the same die in various positions. After observing these figures, we have to find the number opposite a given number on the die. The procedure to be adopted for solving such problems, will be clear from the following examples :

Example 1 : A die is thrown four times and its four different positions are given below. Find the number on the face opposite the face showing 2.



Solution : Here, the number 2 appears in three dice, namely (i), (ii) and (iv). In these dice, we observe that the numbers 4, 1 and 6 appear adjacent to 2. So, none of these numbers can be present opposite 2. The only number left is 5. Hence, 5 is present on the face opposite 2. The answer is (c)

Example 2 : Shown below are, four different positions of the same dice. Find the number on the face opposite the face showing 6.

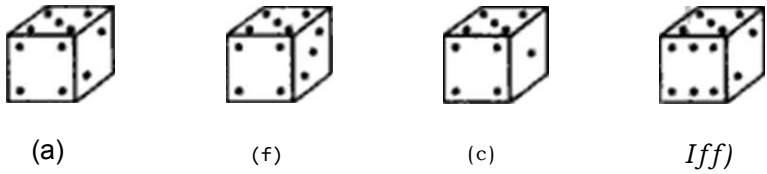


Solution : In this case, the number 6 appears in only two dice from which we observe that the numbers 1, 3 and 5 appear adjacent to 6, so that 2 or 4 can appear opposite 6. So, we begin finding a number which appears at least in three of the given dice. 3 is such a number, which appears in (i), (ii) and (iii). We observe in these dice that, the numbers 1, 4, 5 and 6 appear adjacent to 3. So, they cannot appear opposite 3. The only number that can appear opposite 3 is 2. So, 2 cannot appear opposite 6. Hence, 4 appears opposite 6. so that (c) is the answer

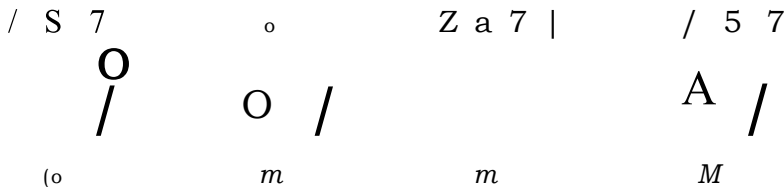
We are now in a position to solve the following exercise.

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6. If the total number of dots on opposite faces of a cubical block is always 7, find the figure which is correct. (Clerk's Oatfo, 1995)



Directions : Questions 7 to 9 are based on the following illustrations, which are four views of a cube. (R*ftwty», 1993)



Study these illustrations carefully and attempt questions 7 to 9.

7. The symbol at the bottom of (iv) is



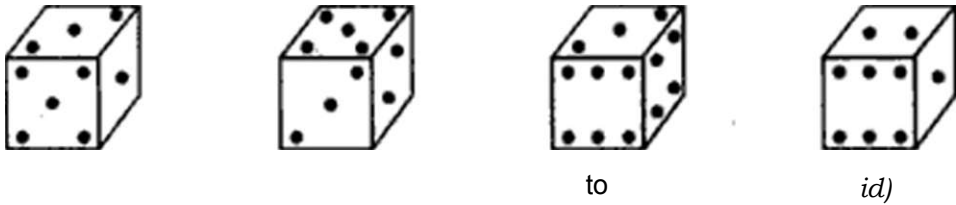
8. The symbol opposite the face having the symbol - is



The symbol opposite the face having triangle is



10. If the total number of dots on opposite faces of a cubical block is always 7, find the figure which is correct. (S.S.C. 1995)



11.

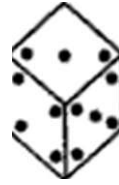


(I-A-S. 1982)

What should be the number opposite 3 7



12.



∞

»

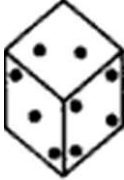
How many dots are opposite 2 dots ?

(a) 1

(b) 3

(c) 6

13.



(i)



(ii)



Find the number of dots on the face opposite the face bearing 3 dots.

(a) 5

(b) 6

(c) 4

(d) Cannot be determined

14.



(i)



(ii)



(iii)

Which numbers occur at the bottom face in the three positions of the same die ?

(a) 6, 6, 2

(b) 5, 6, 1

(c) 0, 5, 5, 5

(d) 6, 5, 2

15.

77) Six faces of a die have been marked with alphabets A, B, C, D, E and F respectively. This die is rolled down three times. The three positions are shown as:



(i)



(ii)



(iii)

Find the alphabet opposite A.

(a) B

(b) C

(c) D

(d) E

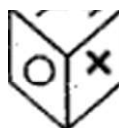
16.



(i)



(ii)



(iii)

A cube has six different symbols drawn over its six faces. The symbols are dot, circle, triangle, square, cross and arrow. Three different positions of the cube are shown in figures (i), (ii) & (iii).

(i) Which symbol is opposite the dot ?

- (a) Circle (b) Triangle (c) Arrow (d) Cross

(ii) Which symbol is opposite the Arrow?

- (a) Circle (b) Triangle (c) Dot (d) Cross

(iii) Which symbol occurs at the bottom of fig. (ii).

- (a) Arrow (b) Triangle (c) Circle (d) Dot

17.



(X)



(Y)



(Z)

(i) Which number lies at the bottom face of the die X ?

- (a) 1 (b) 2 (c) 3 (d) 4

(ii) Which number lies at the bottom face of the die Y ?

- (a) 6 (b) 5 (c) 2 (d) 1

(iii) Which number lies opposite 6?

- (a) 1 (b) 2 (c) 4 (d) 5

(iv) Which numbers are hidden behind the numbers 6 and 5 in the die Z?

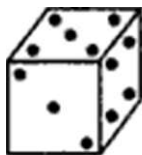
- (a) 1 & 4 (b) 1 & 3 (c) 4 & 3 (d) 1 & 2

(v) Which of the hidden numbers adjacent to 5 in die X are common to the hidden numbers adjacent to 5 in die Z?

- (a) 1 & 4 (b) 2 (c) 6 (d) None

18. Two positions of a dice are shown below. If 1 is at the bottom, which number will be on the top ?

(Awtt Orad0 1994)



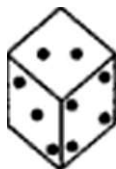
(i)



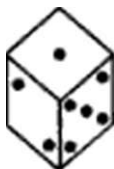
(ii)

- (a) 2 (b) 3 (c) 4 (d) 5

19.



(i)

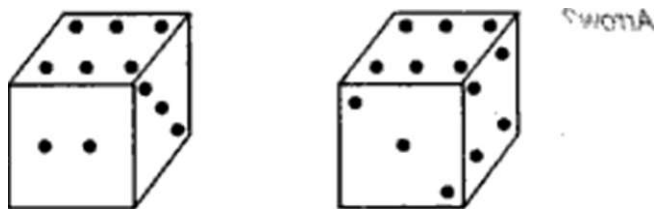


(ii)

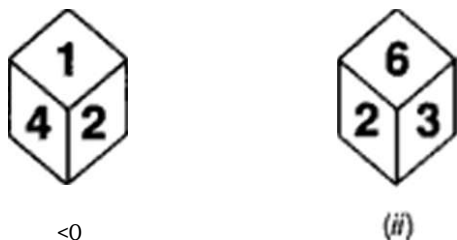
What is the number of dots at the bottom face of the left hand side die ?

- (a) 3 (b) 4 (c) 5 (d) 6

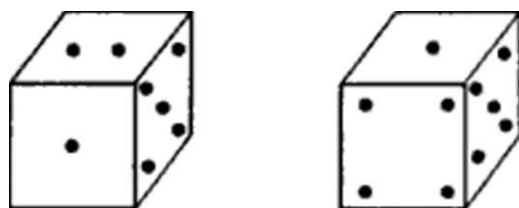
20. Two positions of a dice with 1 to 6 dots on its sides are shown below. If the dice is resting on the side with three dots, what will be the number of dots on the side at the top?
(Section Officers, 1993)



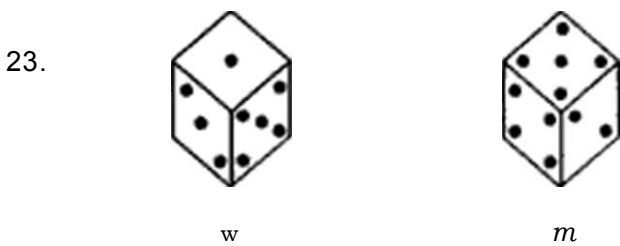
- (a) 1 or 5 (b) 2 (c) 3 (d) 5
21. What will be the number at the bottom if 5 is at the top: the two positions of the dice being as given below :



- (a) 1 (b) 2 (c) 3 (d) 6
22. Observe the dots on a dice (one to six dots) in the following figures. How many dots are contained on the face opposite to that containing four dots?
(I. Tax & Central Excise, 1996)



- (a) 2 (b) 3 (c) 5 (d) 6



23. What is the number of dots on the face opposite 2 dots ?
(a) 1 (b) 3 (c) 4 (d) 6

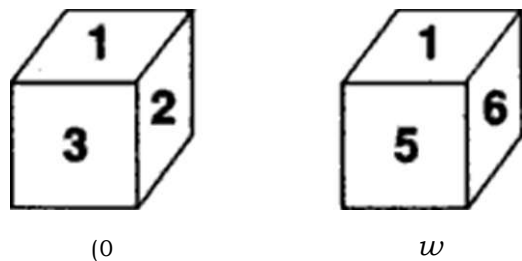
24. Two positions of a block are shown below. When 2 is at the bottom, which number will be at the top?
(U.D.C. 1995)

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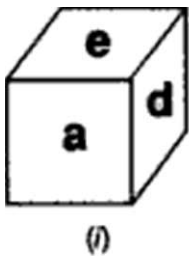
When Blue is on the top. which colour will be at the bottom ?

- (a) Orange **ib)** Red (c) White If f) Yellow

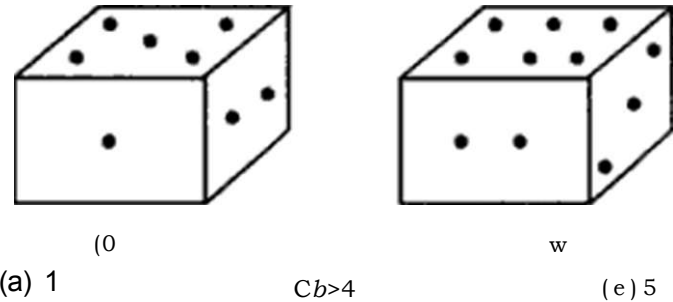
29. Two positions of a dice are shown. When 4 is at the bottom, what number will be on the top ?
(Asstt. Grade, 1995)



30. In a dice a, b, c and d are written on the adjacent faces, in a clockwise order and e and f at the top and bottom. When c is at the top, what will be at the bottom?
(Asstt Grade, 1995)



31. Two positions of a parallelopiped are shown below. When the number 3 will be on the top side, then which number will be at the bottom ?
(L Tax A Central Excise, 1994)



32. Two positions of a dice are shown below. When there are two dots at the bottom, the number of dots at the top will be
(Asstt. Grade, 1995)



33. Two positions of a block are shown below : (U.D.C. 1995)



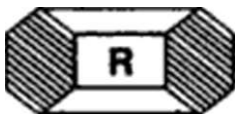
When six is at the bottom, what number will be at the top ?

- (a) 1 (b) 2 (c) 4 (d) 5

34. The lateral sides of a block in the shape of a six-sided prism is painted in Violet, Blue, Green, Yellow, Orange and Red. Two of its positions are shown below :

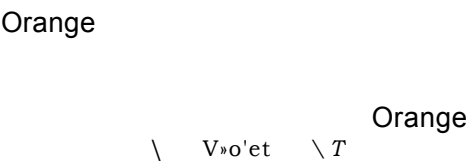


When the block is laid as in the figure what are the colours adjacent to the Red side ? (Asstt. Grade, 1994)



- (a) Yellow and Orange (b) Yellow and Blue
(c) Violet and Yellow (d) Violet and Orange

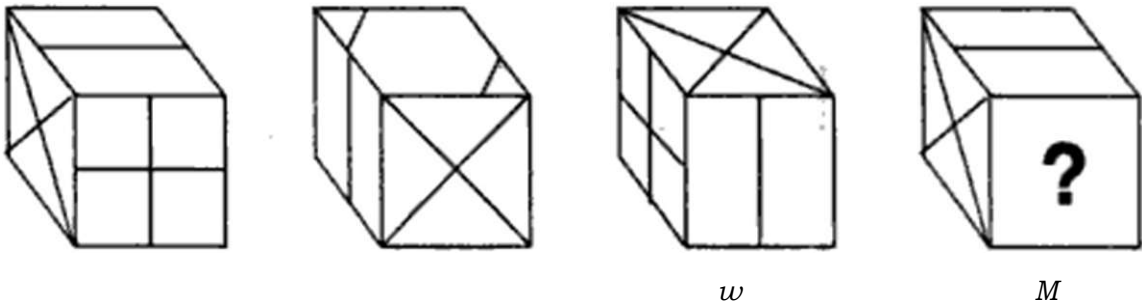
35. A cylinder is painted in 6 colours" Green, Blue, Yellow. Violet. Red and Orange. Two positions are shown below : (Asstt. Grade, 1994)



What is the colour in the empty space ?

- (a) Blue (i) Green (d) Violet (d) Yellow

36. A cubical block with designs in the faces is presented as viewed from different directions. Find the design on the blank face? (Asstt. Grade, 1993^



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ANSWERS

- M a) : From fig. (i), (ii) and (iii), it is clear that the numbers 3, 2, 4 and 5 lie adjacent to the number 6. So, 1 lies opposite 6.
- 2- (a): From fig. (i), (ii) and (iii), it is clear that the numbers 1, 5, 4 and 2 lie adjacent to the number 3. So, 6 lies opposite 3. From fig. (ii) and (iii), it follows that 3, 5 and 2 lie adjacent to 4. So, 1 and 6 can lie opposite 4. But 6 lies opposite 3. Therefore, 1 lies opposite 4.
3. (c) : From fig. (i), (iii) and (iv), it is clear that the numbers 6, 1, 5 and 2 lie adjacent to 3. So, 4 lies opposite 3.
4. (a): From fig. (iii) and (iv), it is clear that the numbers 5, 6, 2, and 3 lie adjacent to 4. So, 1 lies opposite 4.
5. (c): From fig. (i), (iii) and (iv), we find that the numbers adjacent to 4 are 5, 6, 1 and 2. So, 3 lies opposite 4 i.e. 4 lies opposite 3.
8. (c): Since the sum of the number of dots on opposite faces of the block is always 7, we cannot get 1 dot adjacent 6 dots, 2 dots adjacent 5 dots or 3 dots adjacent 4 dots. So, the figures (a), (6) and (d) cannot be correct.
7. (d): From fig. (i), (iii) and (iv), we observe that the symbols O' B0 \ V & lie adjacent to So, 4 £ 3 > lies opposite ^' Q ^.
8. (c) : From fig. (ii) and (iii), it is clear that the symbols ^ • \ ^0' 1_____f and ^A' lie adjacent to V sign. So, O lies opposite V
9. (a): From fig. (iii) and (iv), it follows that the symbols \ V & O' lie adjacent to ^A'. So one of the symbols 'O' or ' • • ' can lie opposite ^A \ But lies opposite ^ 1_____I'. So 'O' lies opposite ^A \
10. (a): Since the sum of the number of dots on opposite faces of the block is always 7, we cannot get 1 dot adjacent 6 dots, 2 dots adjacent 5 dots or 3 dots adjacent 4 dots. So, the figures (6), (c) and (c/) cannot be correct.
11. (6): From the three given figures, it is clear that the numbers 1, 2, 5 and 4 appear adjacent to 3, so none of these can appear opposite 3. Therefore, 6 appears opposite 3.
12. (c): From figures (ii) and (iii), it is clear that one, three, four and six dots cannot appear opposite five dots. Therefore, two dots appear opposite five dots.
13. (6): From fig. (ii) and (iii) it is clear that six, four, one and three dots cannot appear opposite two dots. So, five dots appear opposite two dots. From fig. (i) and (iii) it is clear that four, two and one dots cannot appear opposite three dots. Also, since five dots appear opposite two dots so they cannot appear opposite three dots. Therefore, six dots appear opposite three dots.
14. (c): From three figures it is clear that the numbers 2, 3, 1 and 6 cannot appear opposite 4. So, 5 appears opposite 4. Since in each one of the three dice, 4 appears on the top. So, 5 appears at the bottom face of each dice.
15. (d): From fig. (ii) and (iii) it is clear that C, D, B and F cannot appear opposite E. So, A appears opposite E. i.e. E is the alphabet opposite A.
16. From fig. (i) and (iii), it is clear that dot, triangle, cross and arrow cannot appear opposite the circle. So, the square lies opposite the circle.
From fig. (ii) and (iii), it is clear that triangle, square, arrow and circle cannot appear opposite the cross. So, the dot lies opposite the cross. Obviously the triangle lies opposite the arrow.
- (i) (d): As discussed earlier, the cross lies opposite the dot.
- (ii) (6): As discussed earlier, the triangle lies opposite the arrow.
- (iii) (c): Since the square lies at the top of fig. (ii) and the circle lies opposite the square; so, the circle lies at the bottom of fig. (ii).

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